Code-Layout, Readability and Reusability

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	***OptiCrop: Smart Agricultural Production Optimization Engine***
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	*Code Quality Parameter*	*Description*	*Followed (Yes / No / Partial)*	*Remarks*
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	
2	Proper File Structure	Files and folders are logically organized	Yes	
3	Meaningful Variable Names	Variables clearly represent their purpose	Yes	
4	Function / Method Names	Functions are descriptively named	Yes	
5	Code Comments	Inline and block comments explain logic	Partial	
6	Modular Design	Code is split into reusable functions/ modules	Yes	
7	No Redundant Code	Duplicate or unused code is removed	Yes	
8	Error Handling	Exceptions handled properly in backend and model pipeline	Yes	

Reusable Components / Modules:

S.No	*Component / Module Name*	*Language / Technology*	*where Reused*	*Reusability Level (High / Medium / Low)*
1	**Data Preprocessing Pipeline**	Python, Pandas, Scikit-learn	Training + Prediction	

	(NPK, scaling, encoding)		(Flask App)	***High***
2	**Machine Learning Model** (Crop Prediction Model)	Scikit-learn (KNN, Logistic Regression, Random Forest)	Flask Backend (model.pkl inference)	***High***
3	**Flask Web Application** (API Layer)	Flask, Python	Handles UI input & prediction output	**High**
4	**Dataset Handling & EDA Module**	Pandas, NumPy, Seaborn, Matplotlib	Data analysis & visualization	**Medium**
5	**Utility Functions** (cleaning, feature extraction)	Python	Preprocessing + model pipeline	**Medium**
6	**Model Evaluation Module**	Scikit-learn metrics	Model comparison & accuracy checking	**Medium**

Overall Code Quality Assessment:

Aspect	***Rating (1-5)***	***Comments***
Code Layout & Structure	5	Well-structured project with clear separation of dataset, model, and Flask app components
Readability	4.5	Clean and understandable code with standard Python conventions
Reusability	4.5	Preprocessing pipeline and trained model reused effectively in deployment
Documentation / Comments	5	Good comments and explanations, could improve with more detailed function-level docstrings
Overall Score	4.7	The OptiCrop project is a complete end-to-end machine learning system for crop prediction, covering data preprocessing, model training, evaluation, and Flask-based deployment.